

Samiul Mushfik

(763) 238-2122 | [linkedin.com/in/samiulmushfik](https://www.linkedin.com/in/samiulmushfik) | samiulmushfik123@gmail.com | github.com/samiul123 | Faribault, MN

EDUCATION

University of Minnesota, MS in CS - GPA: 4.0

August 2025, Duluth, MN

Bangladesh University of Engineering and Technology, BS in CSE

April 2019, Dhaka, Bangladesh

PROFESSIONAL EXPERIENCE

Mass Technologies LLC, Software Developer Intern, Remote

April 2026 – Present, Westerville, OH

Healthcare planning provider, improving hospital operations and patient care through technology integration

- Developing an AI-driven pipeline to extract structured installation requirements from manufacturer PDFs for ~1M medical devices, powering dynamic hospital infrastructure checklists.
- Architected an async bulk-import pipeline using AWS Step Functions, Distributed Map, S3, and Lambda, processing 50K-row files concurrently without exhausting workflow state or backend memory.

Stibo DX, Software Engineer

August 2022 – August 2023, Dhaka, Bangladesh

Enterprise headless content management solutions provider, powering content management for 200+ media organizations

- Optimized GraphQL content retrieval, reducing latency by 20% through concurrent data fetching and Node.js clustering.
- Designed a unified CLI and containerized development workflow for 5+ presentation-layer microservices, enabling consistent cross-platform builds and simplifying developer onboarding.

Kona Software Lab Ltd., Software Engineer

May 2019 – July 2022, Dhaka, Bangladesh

Fintech R&D arm of KONAI, present in 90+ countries, and serves 120M+ users

- Shipped a high-throughput transaction-processing API gateway (12K+ RPM) over TCP protocol for point-of-sale (POS) systems, supporting \$2.5B+ in annual financial revenue using Java, Spring Boot, and Apache Camel.
- Designed and implemented POS device authentication protocols and JWT-based session management, enabling encrypted real-time communication through mutual auth, nonce validation, AES-256 symmetric encryption, and Redis.
- Reduced P95 latency to sub-100ms across 1M+ daily financial transactions by optimizing distributed system bottlenecks using Redis caching, async processing, and SQL tuning.
- Designed a Kafka-based event-driven pipeline processing telemetry from 1K+ POS devices, improving real-time system observability and enabling faster detection of production issues.
- Engineered WebSocket infrastructure supporting 100K+ concurrent connections, enabling low-latency administrative operations using Nginx and Redis Pub/Sub.
- Owned production support for 5+ microservices, resolving critical incidents through log analysis and root cause investigation in Kibana, improving system reliability and uptime.
- Automated continuous integration/continuous delivery (CI/CD) failure alerts via Jenkins-Slack integration, cutting build feedback time from hours to minutes across 70+ weekly builds.

Parkspace, Founding Engineer

August 2019 – February 2020, Dhaka, Bangladesh

Pre-launch mobility startup building a real-time parking discovery and reservation app

- Implemented core features such as park finding, reservation, cancellation, etc., for a real-time parking discovery and reservation MVP (React Native, Spring Boot, and AWS).
- Maintained 80% test coverage through TDD and streamlined deployments using CircleCI-based CI/CD pipeline.

RESEARCH EXPERIENCE

University of Minnesota, Graduate Student Researcher

August 2023 – August 2025, Duluth, MN

- Optimized large-scale in-memory R-Tree construction using AVX-512 SIMD, achieving over 3x speedup on 70M records by improving memory access and parallel computation efficiency.

OPEN-SOURCE CONTRIBUTIONS

- **Monocle** Designed and implemented a PostgreSQL span exporter to enable robust archiving and querying of OpenTelemetry traces directly within the RDBMS.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, C++

Frameworks & APIs: Spring Boot, Node.js, FastAPI, React, React Native, REST, GraphQL, JWT

Data & Distributed Systems: MySQL, PostgreSQL, Redis, Kafka

Cloud, DevOps, Observability & Testing: AWS (EC2, S3, Lambda, IAM, CDK), Docker, Jenkins, Git, Kibana, JUnit, Jasmine

AI Development Tools: Claude Code, GitHub Copilot

CERTIFICATIONS

AWS Certified Developer – Associate

April 2025